

ADHD OS: System Design

Capture it → break it down → do one step → with a timer and company

Status: **Approved** (monorepo, Google + email/password sign-in, names kept) · Date: 2026-10-08 · Revision 2: **\$0 budget. The app works fully without AI, and AI is an optional free add-on**

1. The plain-English version

What it is

ADHD OS is a daily home base for getting things done when your brain doesn't cooperate. It puts four tools into one simple loop:

1. **Dump it.** Get the thought out of your head (this reuses the **Brain Dump** app).
2. **Shrink it.** Pick one task and break it into tiny steps, then see **only the next one**. "Clean apartment" becomes "Put the 3 mugs on your desk in the sink." The app helps with **ready-made step templates** and **guiding questions**. If you turn on the optional AI switch, it can write the steps for you.
3. **Do it.** A **visual timer** (a shrinking colored disc, not scary numbers) runs while you do that one step. You get a gentle chime halfway through and near the end.
4. **Don't do it alone.** Join a **body doubling room**: other people (friends, or anyone online) working quietly at the same time. You post your goal when you join and tick it off when you're done.

Why it helps with ADHD

ADHD struggle	What ADHD OS does
"I can't start" (task paralysis)	Steps so small they feel silly, and only one shown at a time
Time blindness	Time you can see shrinking
Overwhelm from long lists	No long list on the main screen. Just one thing
Motivation drops when alone	Other people's presence (body doubling)
Forgetting ideas mid-task	One-tap dump, always one button away

What does it cost?

Nothing. Every piece runs on a free plan. The timer, rooms, templates and everything else work without AI. The AI switch is optional and uses a free service.

How it works, as an analogy

Imagine a **coach** sitting next to you:

- You hand them a messy pile of notes (**Brain Dump**).

- You point at one note. They pull out a **recipe card** for that kind of job, or ask "*what's the very first thing you'd physically touch?*", and hide the rest (**Task Breaker**).
- They put an **hourglass** on the table (**Visual Timer**).
- And you're sitting in a **quiet library** where everyone else is working too (**Body Doubling**).

Behind the scenes it's the same "office" as Brain Dump: Vercel hosts the website and a database keeps your tasks. One new piece: a **live connection service** that lets the app show who is in the room right now.

2. Features (version 1)

Home screen: "The One Thing"

- Shows a single card: your current step, a **Start** button and a **Dump** button. Nothing else.
- "Not this one" button skips to a different task without guilt.

Task Breaker (no AI needed)

- Pick any task (from Brain Dump or typed fresh).
- **Templates:** a built-in library of step lists for common ADHD-hard tasks, e.g.:
 - *Clean a room · Do laundry · Reply to a hard email · Pay a bill*
 - *Study for an exam · Cook a simple meal · Get ready to leave the house · Make a phone call*
 - Pick one → its steps are copied in → edit freely.
- **Guided breakdown:** if no template fits, the app asks short questions one at a time:
 1. "*What does 'done' look like?*"
 2. "*What's the very first thing you'd physically touch?*"
 3. "*And after that?*" (repeat)
- **Too big?** on any step asks "*What's the smallest piece of this?*" and splits it.
- **Save as template:** turn any finished breakdown into your own reusable template.
- Each step gets an optional time estimate (2 / 5 / 10 / 15 min chips).
- Shows one step at a time. **Done** reveals the next.

AI assist (optional switch, off by default)

- When on, **Break it down for me** sends the task title to a **free-tier AI service**, which returns 3–10 tiny, concrete steps with time estimates. You can edit everything.
- **Too big?** can also ask the AI to split a step.
- If the free limit is hit or the service is down, the app falls back to templates and guided questions.

Visual Timer

- Shrinking disc with a color shift (calm blue → warm orange near the end).
- Presets: 5 ("just 5 minutes"), 15, 25 minutes, or the step's estimate.
- Soft chimes at 50% and with 1 minute left. Optional vibration on phones.
- When it ends: "Keep going?", "Done!" or "Break". No shame either way.
- Runs in the browser and keeps the right time even if you switch tabs (it uses the start timestamp, not a ticking counter).

Body Doubling Rooms

- **Public room:** anyone signed in can drop in.
- **Private rooms:** share a link with friends.
- When you join, type your goal ("finish the email to my landlord").
- Everyone sees live presence cards: name, goal, timer ring, and a ✓ when done.
- Optional video: a "Turn on video" button opens a free Jitsi Meet call for that room (no extra accounts).
- Quiet by design: no chat feed, only short preset reactions (👏 🎉 🙌).

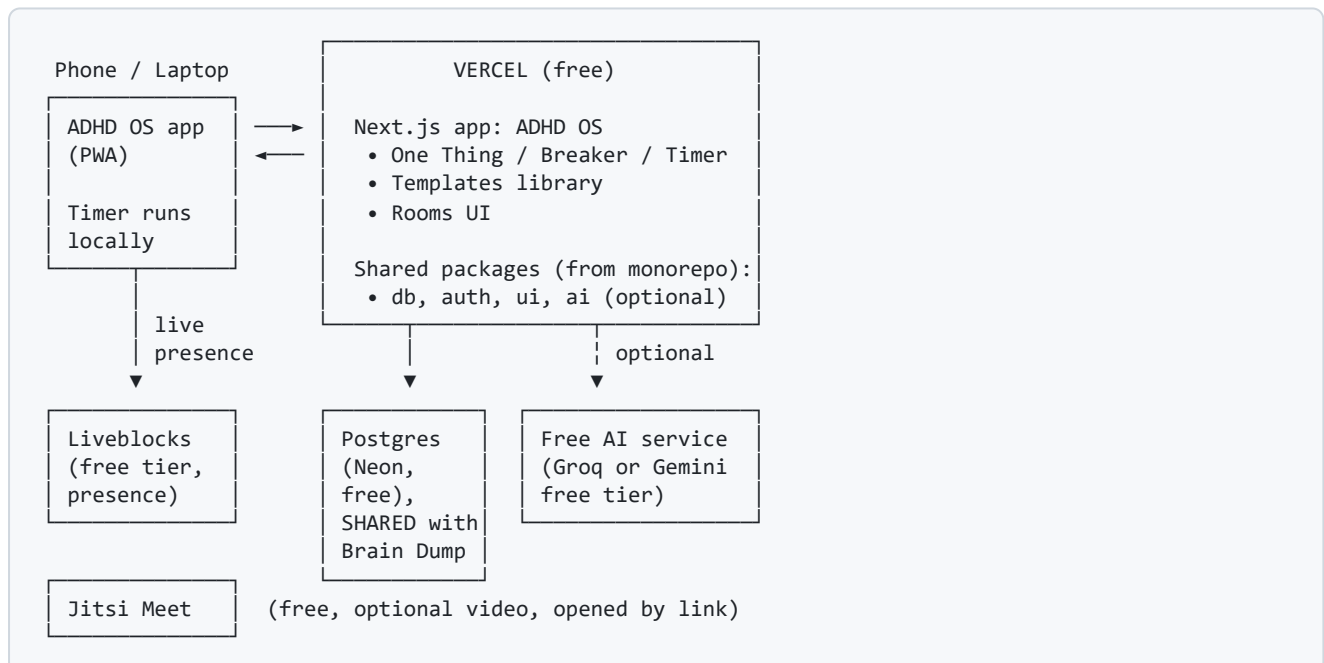
Light gamification

- Daily "steps done" count and a **forgiving streak** (missing one day doesn't reset it).
- Small celebration animation on each completed step.

Later (version 2)

- Absorb more of your backlog as modules: Overwhelm button, Dopamine menu, "Leave by" calculator, Time estimate trainer (estimated vs. actual times are already recorded, so this is mostly a new screen).
- Push notifications ("Your friend just joined the room"), via free Web Push.

3. Architecture



Why a separate realtime service?

Vercel runs code in short bursts (serverless functions). It's great for "load a page" or "save a task", but it can't hold a connection open for hours to stream "who's in the room right now". **Liveblocks** is built exactly

for live presence and has a free tier. The app asks our server for a room token (so only signed-in users can join), then connects straight to Liveblocks.

Tech choices (all free)

Layer	Choice	Why
Framework	Next.js + TypeScript , same as Brain Dump	One skill set, shared code
Repo layout	Turborepo monorepo : <code>apps/brain-dump</code> , <code>apps/adhd-os</code> , <code>packages/*</code>	Write sign-in, database, AI and UI code once, use in both apps
Database	Same Neon Postgres as Brain Dump	Something dumped in one app appears in the other
Auth	Better Auth , same Google + email/password sign-in	One account across both apps (matched by email)
Task breaking	Templates (JSON in code) + guided questions	Free, private, always works
AI assist (optional)	Vercel AI SDK + free-tier provider (Groq or Gemini)	Free up to daily limits; provider swappable in one line
Realtime	Liveblocks free tier (presence only)	Holds live connections, which Vercel can't
Video	Jitsi Meet link (optional)	Free, no setup, no accounts
Timer	Pure browser (timestamps + Web Audio chimes)	Accurate, works offline, no server cost

4. Data model (adds to the Brain Dump tables)

```
items (from Brain Dump; a task lives here)
  id, user_id, kind, title, ... , focus_order?  ← new: position in "One Thing" queue

task_steps      micro-steps for one task
  id, item_id, user_id, position, text,
  est_minutes?, source (template|guided|manual|ai),
  done_at?, skipped_at?, created_at

templates      user-saved templates (built-in ones live in code)
  id, user_id, name, steps (json), created_at

focus_sessions one per timer run
  id, user_id, step_id?, planned_minutes,
  started_at, ended_at?, outcome (done|continued|break|abandoned)

rooms
  id, owner_id?, name, is_public, invite_code, created_at

room_sessions  who joined which room, with what goal
  id, room_id, user_id, goal, joined_at, left_at?, completed (bool)

streaks        cached daily totals for fast display
  user_id, date, steps_done
```

Live "who's here right now" lives in Liveblocks (temporary). `room_sessions` keeps the permanent history ("You've done 14 body-doubling sessions this month").

5. Key flows

A. Break down a task (no AI)

1. On a task, tap **Break it down**.
2. The app suggests matching templates (by keywords in the title, e.g. "laundry" → *Do laundry*).
3. Pick a template → steps copied in. Or choose **Guide me** → answer short questions → each answer becomes a step.
4. Steps are saved to `task_steps`. The home screen now shows step 1.

A2. Break down a task (AI assist on)

1. Tap **Break it down for me** → the server asks the free AI service for 3–10 tiny steps.
2. Steps appear for review → edit → save. On any error, it offers templates/guide instead.

B. Do one step

1. Tap **Start** → timer begins (length = step estimate, or your preset) → `focus_sessions` row created.
2. Chimes at 50% and 1 min left.
3. Tap **Done** → step marked done, session closed, celebration, next step shown.

C. Body double

1. Open **Rooms** → pick public or a friend's link → type goal → Join.
2. The app asks our server for a Liveblocks token (checks you're signed in and allowed in that room).

3. Your presence card appears for everyone instantly. Your timer ring syncs.
4. Tick ✓ when done; leaving closes your `room_session`.

6. Privacy and safety

- Same rules as Brain Dump: every query is filtered to the signed-in user.
- **AI assist is off by default.** With it off, task text never leaves the app's own servers.
- Free AI tiers may use what you send to improve their models; check the provider's terms before turning it on.
- In rooms, others see only **your display name, goal text and timer**, never your tasks or dumps.
- Public room: report/block buttons, short goal length limit and a basic word filter. Private rooms are invite-link only, and links can be regenerated.
- Option to appear as "Anonymous Focuser" in public rooms.

7. Costs

\$0. Vercel Hobby, Neon, Liveblocks and the free AI tier (optional) are all free at personal/friends scale. Jitsi is free.

8. Risks and open questions

Risk / question	Plan
Templates don't cover a task	Guided questions work for anything; save your own templates
Free AI tier changes or disappears	App never depends on it; swappable provider
Public room moderation	Start with private rooms + one public room; add moderation if it grows
Liveblocks free tier limits	Fine for personal/friends use; alternatives (Ably, Pusher, PartyKit) are easy swaps since presence code is isolated
Notifications on iPhone	Needs PWA install; v2

9. Build milestones

(Starts after Brain Dump milestones 1–3, so the monorepo, sign-in and database already exist.)

1. ADHD OS app shell in the monorepo, shared sign-in, deploy as its own Vercel project.
2. Task Breaker: templates library, guided questions, Too big?, save-as-template.
3. "One Thing" home screen.

4. Visual Timer + focus sessions.
5. Body doubling rooms (Liveblocks presence, goals, ✓).
6. Optional Jitsi video button.
7. Optional AI assist for task breaking.
8. Streaks, celebrations, polish.